Continuous-Time System Analysis Using The Laplace Transform

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Outline

- Laplace Transform
- Properties of Laplace Transform
- Solution of Differential Equations
- Analysis of Electrical Networks
- Block Diagrams and System Realization
- Frequency Response of an LTIC System
- Filter Design by Placement of Poles and Zeros of $H(s)$

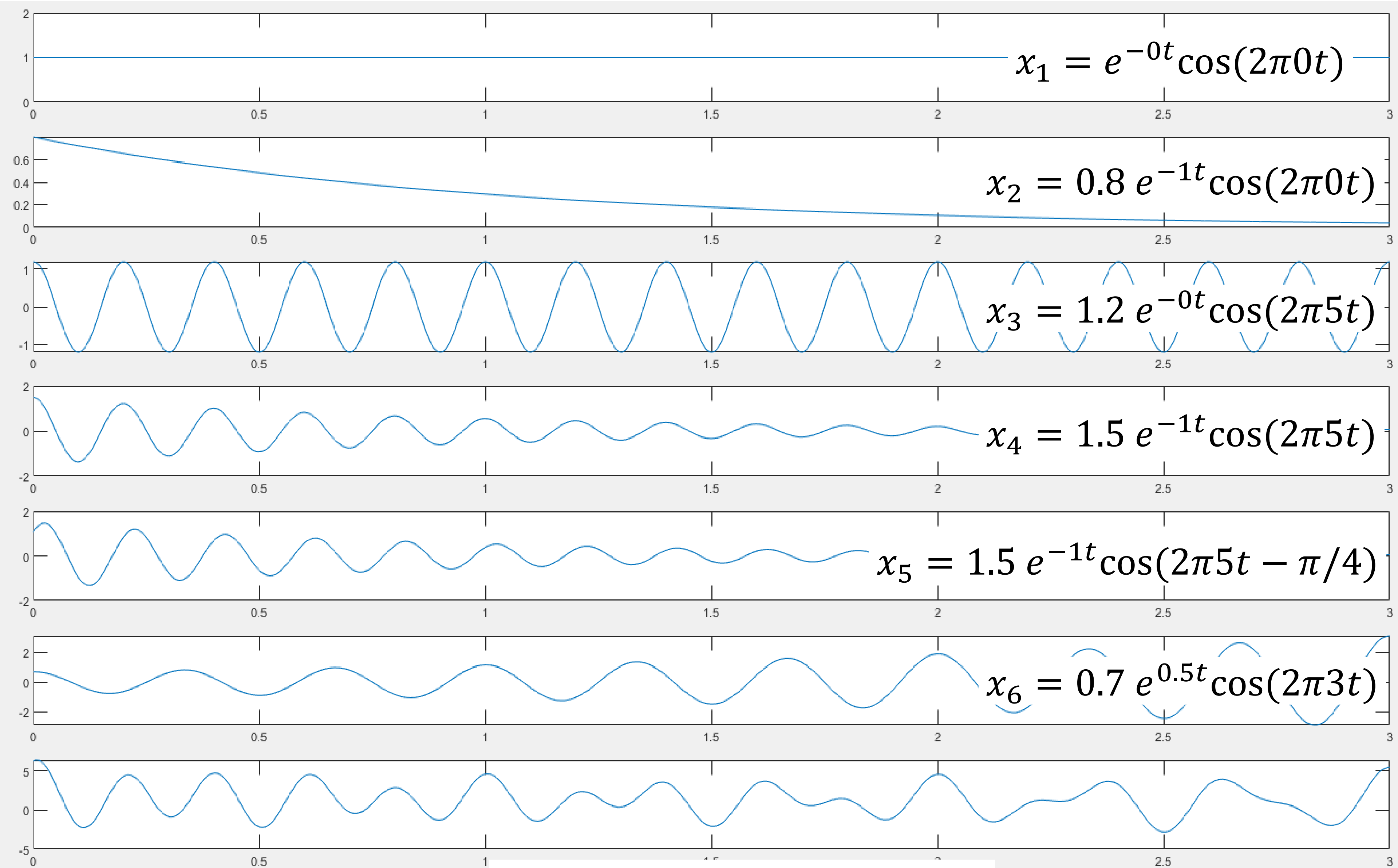
$$x(t) \xleftrightarrow{\text{LT}} X(s)$$

$$x(t) = \sum_i X(s_i) e^{s_i t}$$

s_i : complex frequency = $\sigma_i + j\omega_i$

σ_i : rate of decay

ω_i : rate of oscillation



$$x(t) = x_1 + x_2 + x_3 + \cdots + x_6$$

Why Laplace Transform?

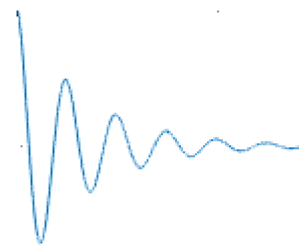
Laplace transform is a mathematical tool to map signals and system behavior from the time-domain into the frequency domain.

Time Domain

$$x(t) = \sum_{n=0} x(nT)\delta(t - nT)$$

Laplace Domain (frequency)

$$x(t) = \sum_{i} X(s_i)e^{s_i t}$$



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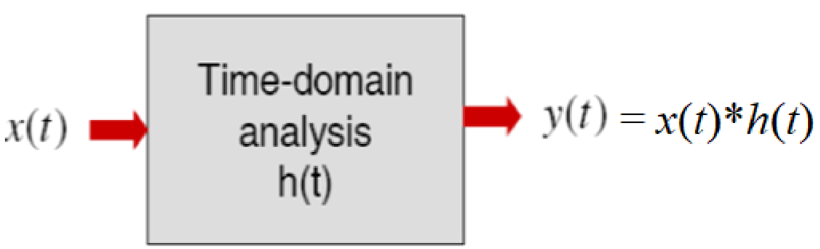
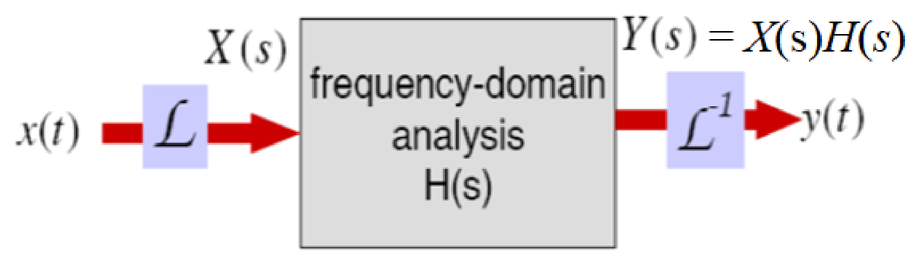




Definition of Laplace Transform

Two-sided Laplace transform of $x(t)$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

One-sided (unilateral) Laplace transform of causal signal $x(t)$

$$X(s) = \int_{0}^{\infty} x(t) e^{-st} dt$$

Two-sided (bilateral) inverse Laplace transform of $X(s)$

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s) e^{st} ds$$

Examples of Popular Functions

Find the Laplace transform of the impulse $\delta(t)$.

$$\mathcal{L}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = 1$$

For all s

Find the Laplace transform of the unit step $u(t)$.

$$\mathcal{L}[u(t)] = \int_{0}^{\infty} u(t) e^{-st} dt$$

$$= -\frac{1}{s} e^{-st} \Big|_{0}^{\infty} = \frac{1}{s}$$

For Re[s]
$\sigma > 0$

Examples of Popular Functions

Find the Laplace transform of $e^{at}u(t)$.

$$\mathcal{L}[e^{at}u(t)] = \int_0^\infty e^{at}e^{-st}dt = \frac{1}{s-a}$$

For Re[s] > a

$\sigma > \mathrm{a}$

Find the Laplace transform of $\cos\omega_0 t\, u(t)$.

$$\mathcal{L}[\cos\omega_0 t\, u(t)] = \int_0^\infty \frac{1}{2}\left[e^{j\omega_0 t} + e^{-j\omega_0 t}\right]e^{-st}dt$$

$$= \frac{s}{s^2+\omega_0^2}$$

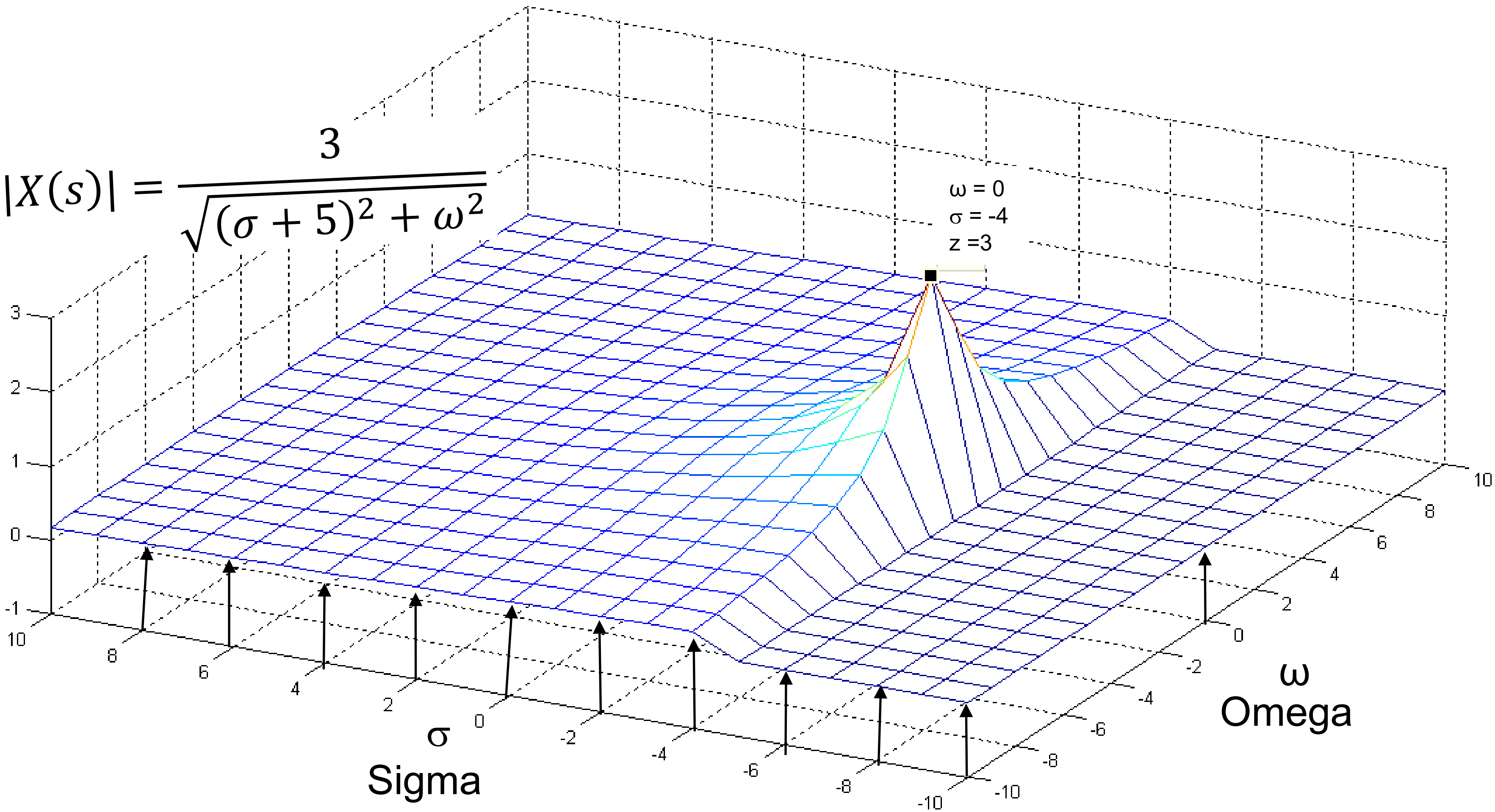
For Re[s] > 0

$\sigma > 0$

Laplace Transform

$$x(t) = 3e^{-5t}\,u(t) \Rightarrow \boxed{\int_{-\infty}^{\infty} x(t)e^{-st}\,dt} \Rightarrow X(s) = \frac{3}{s+5}$$

$$|X(s)| = \frac{3}{\sqrt{(\sigma+5)^2 + \omega^2}}$$



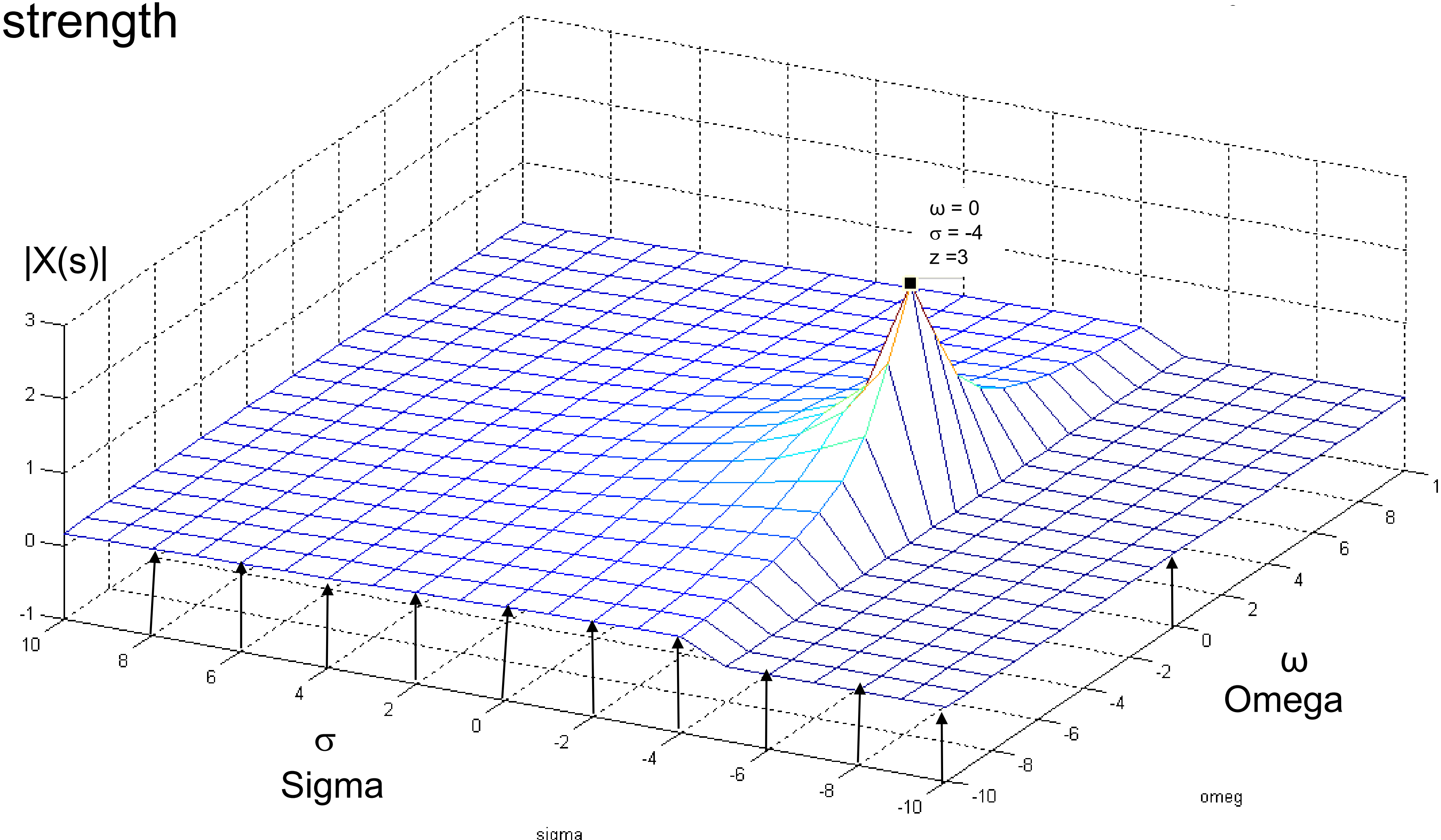
$$X(s) = \frac{6}{s} \Rightarrow \boxed{H(s) = \frac{s+34}{s^2+10s+34}} \Rightarrow Y(s)$$

Inverse Laplace Transform
By Partial Fraction Expansion & the Table

$$x(t) = 6u(t) \qquad h(t) = 10e^{-5t}\cos(3t + 127^{\circ})u(t) \qquad y(t)$$

$$x(t) = 3e^{-5t} \xrightarrow{LT} X(s) = \frac{3}{s+5} \qquad |X(s)| = \frac{3}{\sqrt{(\sigma+5)^2 + \omega^2}}$$

$x(t)$ can be expressed as a linear sum of exponential with different strength



Inverse Laplace Transform

The inverse Laplace transform, finding $x(t)$ from $X(s)$, involves integration in the complex plane that is not covered in this class.

Partial fractions expansion, Laplace properties, and the table will be used to find the inverse Laplace transform.

No.	$x(t)$	$X(s)$
1	$\delta(t)$	1
2	$u(t)$	$\frac{1}{s}$
3	$tu(t)$	$\frac{1}{s^2}$
4	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5	$e^{\lambda t} u(t)$	$\frac{1}{s-\lambda}$
6	$te^{\lambda t} u(t)$	$\frac{1}{(s-\lambda)^2}$
7	$t^n e^{\lambda t} u(t)$	$\frac{n!}{(s-\lambda)^{n+1}}$
8a	$\cos bt\, u(t)$	$\frac{s}{s^2+b^2}$
8b	$\sin bt\, u(t)$	$\frac{b}{s^2+b^2}$
9a	$e^{-at} \cos bt\, u(t)$	$\frac{s+a}{(s+a)^2+b^2}$
9b	$e^{-at} \sin bt\, u(t)$	$\frac{b}{(s+a)^2+b^2}$
10a	$re^{-at} \cos(bt+\theta)\, u(t)$	$\frac{(r\cos\theta)s+(ar\cos\theta-br\sin\theta)}{s^2+2as+(a^2+b^2)}$
10b	$re^{-at} \cos(bt+\theta)\, u(t)$	$\frac{0.5re^{j\theta}}{s+a-jb}+\frac{0.5re^{-j\theta}}{s+a+jb}$
10c	$re^{-at} \cos(bt+\theta)\, u(t)$; $r=\sqrt{\frac{A^2c+B^2-2ABa}{c-a^2}}$; $\theta=\tan^{-1}\left(\frac{Aa-B}{A\sqrt{c-a^2}}\right)$; $b=\sqrt{c-a^2}$	$\frac{As+B}{s^2+2as+c}$
10d	$e^{-at}\left[A\cos bt+\frac{B-Aa}{b}\sin bt\right]u(t)$; $b=\sqrt{c-a^2}$	$\frac{As+B}{s^2+2as+c}$

Example 1 of Inverse Laplace Transform

Use partial fraction expansion to find the inverse Laplace transform of

$$X(s) = \frac{7s + 6}{s^2 + 5s + 6}$$

Answer: $x(t) = [-8e^{-2t} + 15e^{-3t}]u(t)$

Example 2 of Inverse Laplace Transform

Use partial fraction expansion to find the inverse Laplace transform of

$$H(s) = \frac{2s^2 + 5}{s^2 + 3s + 2}$$

Answer: $h(t) = 2\delta(t) + [7e^{-t} - 13e^{-2t}]u(t)$

Example 3 of Inverse Laplace Transform

Use partial fraction expansion to find the inverse Laplace transform of

$$X(s) = \frac{6(s+34)}{s(s^2+10s+34)}$$

$$X(s) = \frac{k_1}{s} + \frac{k_2}{s+5-j3} + \frac{k_3}{s+5+j3}$$

$$X(s) = \frac{k_1}{s} + \frac{As+B}{s^2+10s+34}$$

Answer: $x(t) = \left[6 + 10e^{-5t}\cos(3t + 126.9^{o})\right]u(t)$

Example 4 of Inverse Laplace Transform

Use partial fraction expansion to find the inverse Laplace transform of

$$H(s) = \frac{8s + 10}{(s+1)(s+2)^2}$$

$$H(s) = \frac{k_1}{(s+1)} + \frac{k_2}{(s+2)} + \frac{k_3}{(s+2)^2}$$

Answer: $h(t) = [2e^{-t} - 2e^{-2t} + 6te^{-2t}]u(t)$

Properties of The Laplace Transform

Linearity Property

Laplace Transform

$$x(t) \Rightarrow \boxed{\int_{-\infty}^{\infty} x(t)e^{-st}\,dt} \Rightarrow X(s)$$

$$x(t) \rightleftharpoons X(s)$$

If $\quad x_1(t) \rightleftharpoons X_1(s)$ and $x_2(t) \rightleftharpoons X_2(s)$

then

$$a\,x_1(t) \rightleftharpoons a\,X_1(s)$$

$$a\,x_1(t) + b\,x_2(t) \rightleftharpoons a\,X_1(s) + b\,X_2(s)$$

Time Shifting Property

$$x(t) \rightleftarrows X(s)$$

$$x(t - t_0) \rightleftarrows X(s)e^{-t_0 s}$$

Time delaying $x(t)$ by t_o is equivalent to multiplying *X*(*s*) by $e^{-t_0 s}$ (phase shifting).

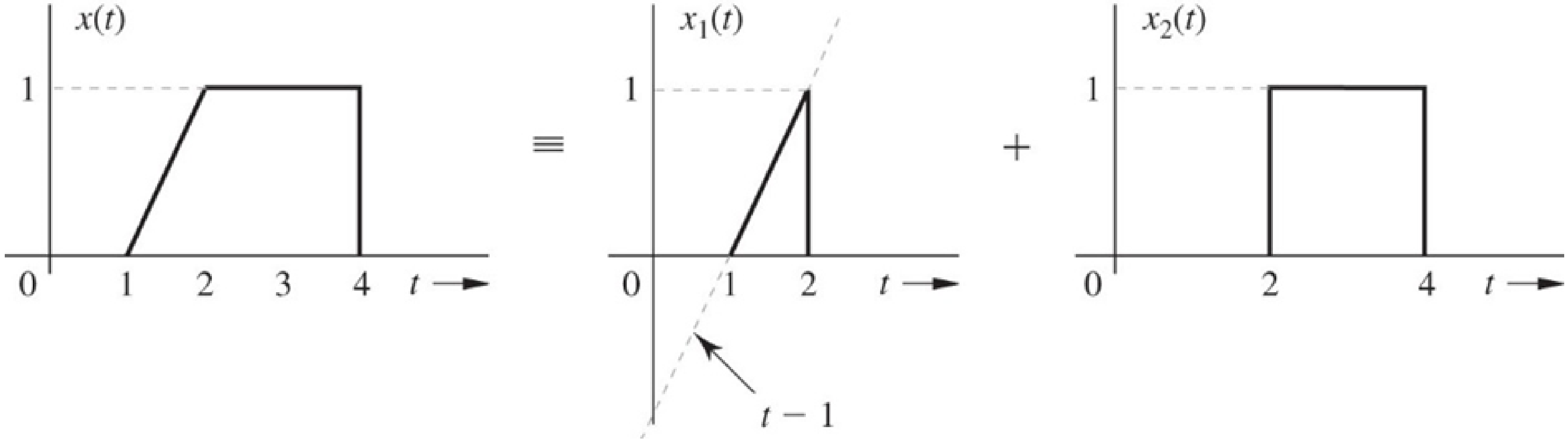
For causal signal

$$x(t)u(t) \rightleftarrows X(s)$$

$$x(t - t_0)u(\mathrm{t} - t_0) \rightleftarrows X(s)e^{-t_0 s}$$

Example: Time Shifting Property

Find the Laplace transform of the signal $x(t)$



Answer:

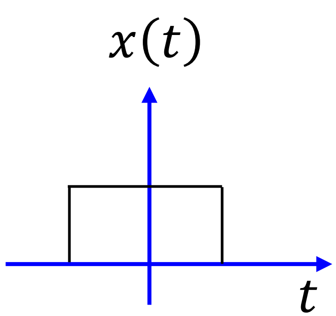
$$X(s) = \frac{1}{s^2}e^{-s} - \frac{1}{s^2}e^{-2s}\frac{1}{s}e^{-4s}$$

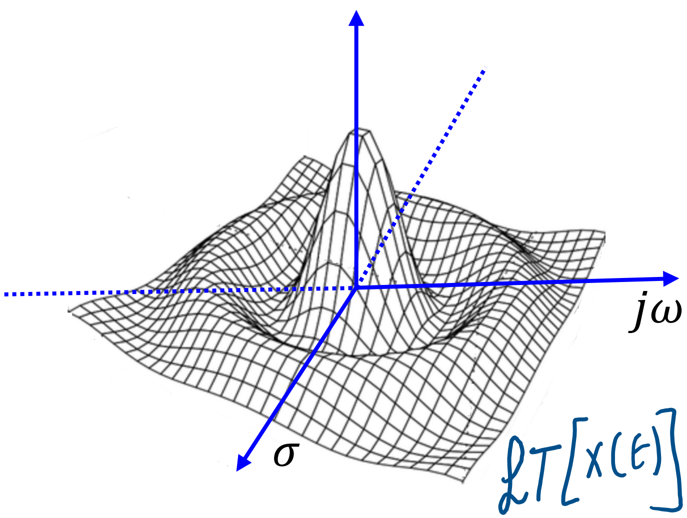
Frequency Shifting Property

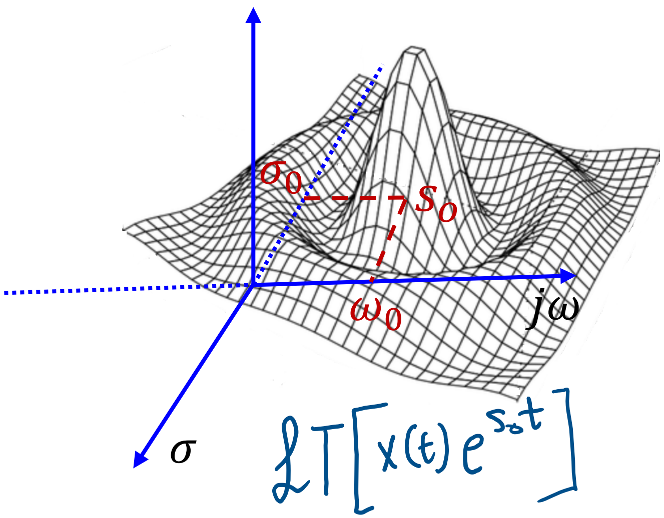
$$x(t) \rightleftarrows X(s)$$

$$x(t)e^{s_o t} \rightleftarrows X(s - s_o)$$

Multiplying the signal $x(t)$ by $e^{s_0 t}$ is equivalent to frequency shifting by s_o in the Laplace domain.







Example: Frequency Shifting Property

Find the Laplace transform of $e^{-at}\cos(bt)\,u(t)$

Answer:

$$X(s) = \frac{s + a}{(s + a)^2 + b^2}$$

Time-Differentiation Property

$$x(t) \rightleftharpoons X(s)$$

$$\frac{dx}{dt} \rightleftharpoons sX(s) - x(0^-)$$

$$\frac{d^2x}{dt^2} \rightleftharpoons s^2X(s) - sx(0^-) - \dot{x}(0^-)$$

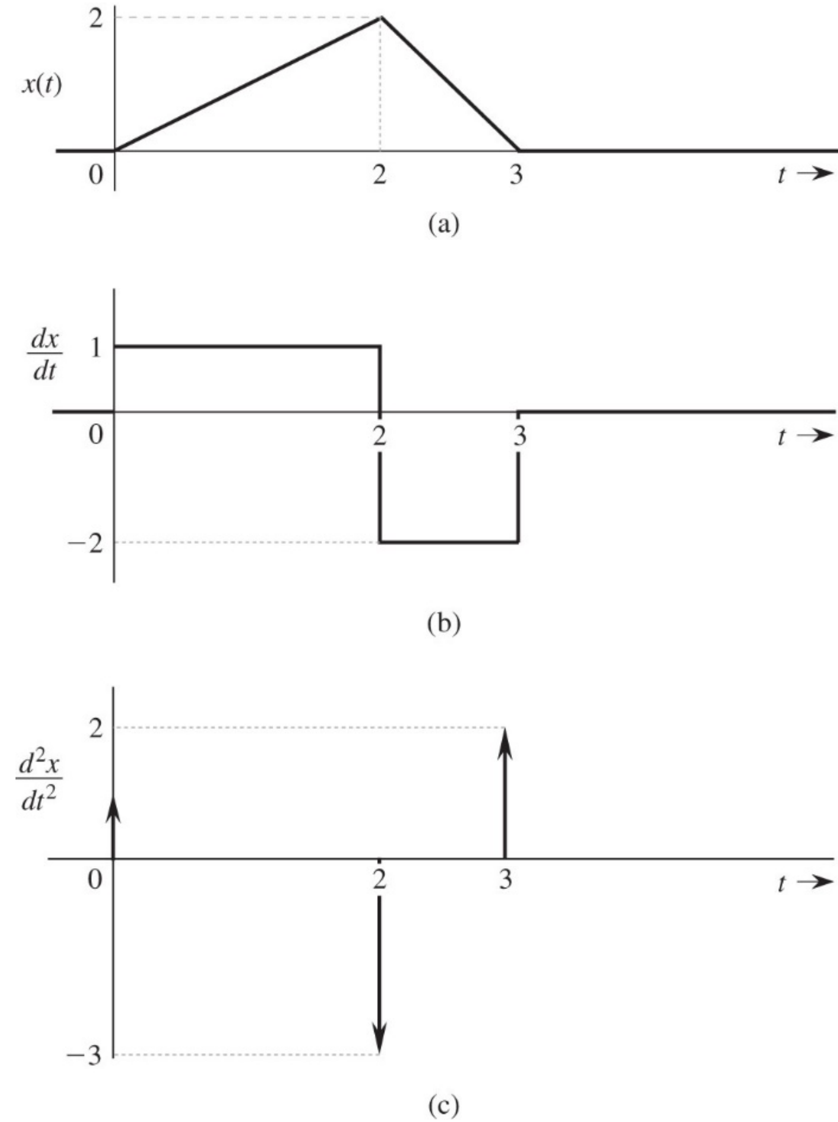
$$\frac{d^3x}{dt^3} \rightleftharpoons s^3X(s) - s^2x(0^-) - s\dot{x}(0^-) - \ddot{x}\,(0^-)$$

$$\frac{d^nx}{dt^n} \rightleftharpoons s^nX(s) - \sum_{k=1}^{n} s^{n-k}\dot{x}^{k-1}(0^-)$$

Frequency-differentiation property $\quad tx(t) \rightleftharpoons -\dfrac{d}{ds}X(s)$

Example: Time Differentiation Property

Find the Laplace transform of the signal $x(t)$



Answer:

$$X(s) = \frac{1}{s^2}\left(1 - 3e^{-2s} + 2e^{-3s}\right)$$

Time-Integration Property

$$x(t) \rightleftharpoons X(s)$$

$$\int_{0^-}^{t} x(\tau)d\tau \rightleftharpoons \frac{X(s)}{s}$$

The dual property of time-integration property is the frequency-integration.

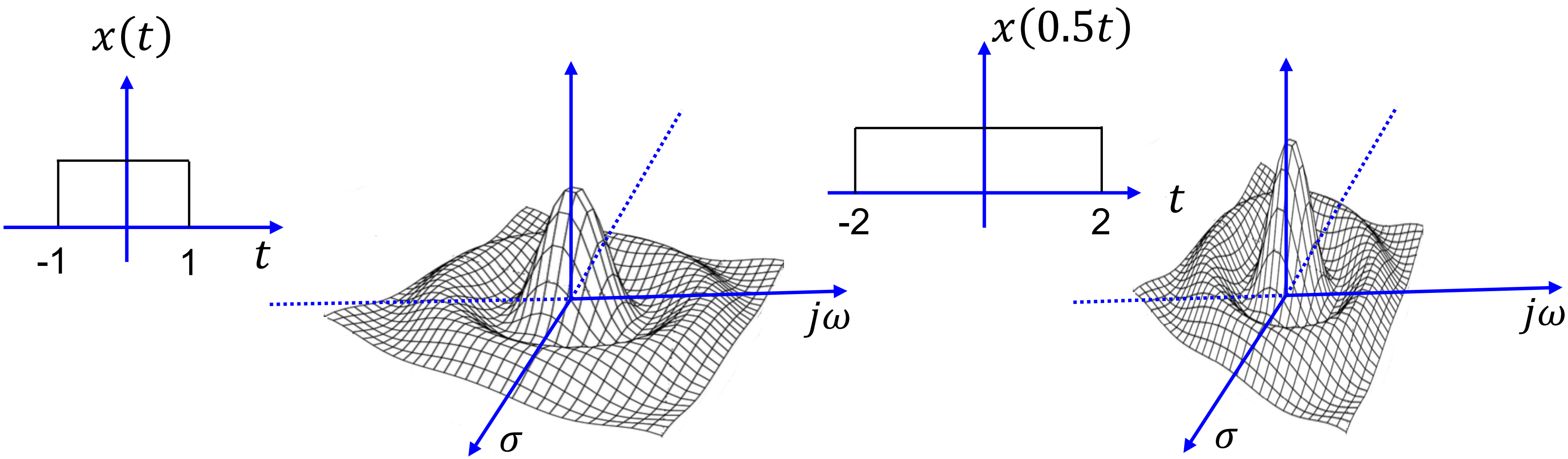
$$\frac{x(t)}{t} \rightleftharpoons \int_{s}^{\infty} X(z)dz$$

Scaling Property

$$x(t) \rightleftharpoons X(s)$$

$$x(at) \rightleftharpoons \frac{1}{a}X\left(\frac{s}{a}\right) \qquad \text{for a} > 0$$

Expansion in time domain result in compression in the s (frequency) domain, and the inverse is true.



Convolution Property

Convolution in the time domain is equivalent to multiplication in the Laplace (frequency) domain.

$$x(t) \rightleftarrows X(s) \qquad h(t) \rightleftarrows H(s)$$

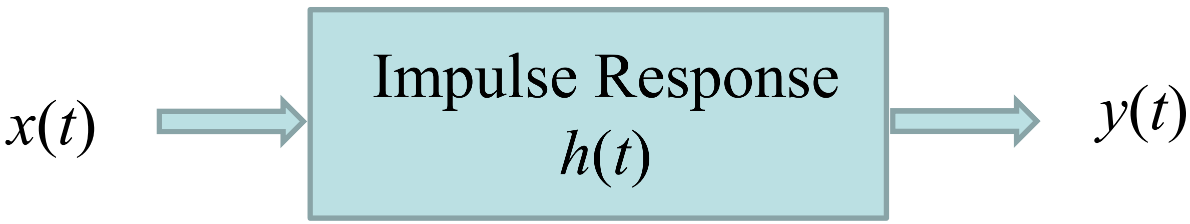
$$x(t) * h(t) \rightleftarrows X(s)H(s)$$

Multiplication in the time domain is equivalent to multiplication in the Laplace (frequency) domain.

$$x(t)h(t) \rightleftarrows \frac{1}{2\pi j}[X(s) * H(s)]$$

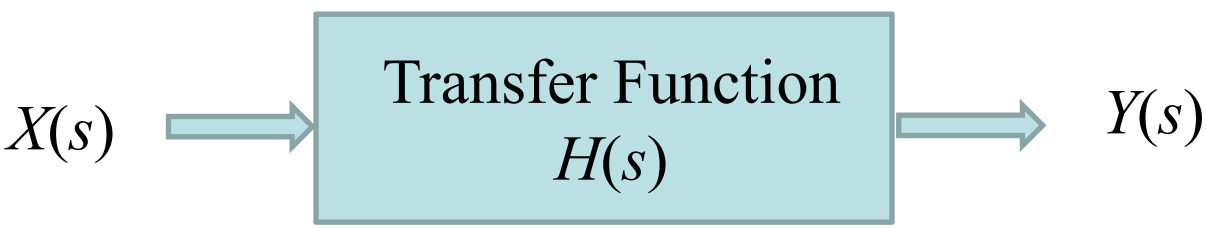
Laplace and Time Domains analysis of Systems

The convolution property is very useful in simplifying the system analysis in the Laplace domain.



$$y(t) = x(t) * h(t)$$

Zero-state response. All initial conditions are zeros



$$Y(s) = X(s)H(s)$$

$$H(s) = \frac{Y(s)}{X(s)}$$

Example: Convolution Property

Find $y(t)$ where $y(t) = e^{at}u(t) * e^{bt}u(t)$

Answer: $y(t) = \frac{1}{a-b}[e^{at} - e^{bt}]u(t)$

Summary of Laplace Transform Properties

Operation	$x(t)$	$X(s)$
Addition	$x_1(t) + x_2(t)$	$X_1(s) + X_2(s)$
Scalar multiplication	$kx(t)$	$kX(s)$
Time differentiation	$\frac{dx}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x}{dt^2}$	$s^2X(s) - sx(0^-) - \dot{x}(0^-)$
	$\frac{d^3x}{dt^3}$	$s^3X(s) - s^2x(0^-) - s\dot{x}(0^-) - \ddot{x}(0^-)$
	$\frac{d^nx}{dt^n}$	$s^nX(s) - \sum_{k=1}^{n} s^{n-k}x^{(k-1)}(0^-)$
Time integration	$\int_{0^-}^{t} x(\tau)\,d\tau$	$\frac{1}{s}X(s)$
	$\int_{-\infty}^{t} x(\tau)\,d\tau$	$\frac{1}{s}X(s) + \frac{1}{s}\int_{-\infty}^{0^-} x(t)\,dt$
Time shifting	$x(t-t_0)u(t-t_0)$	$X(s)e^{-st_0} \quad t_0 \geq 0$
Frequency shifting	$x(t)e^{s_0t}$	$X(s-s_0)$
Frequency differentiation	$-tx(t)$	$\frac{dX(s)}{ds}$
Frequency integration	$\frac{x(t)}{t}$	$\int_{s}^{\infty} X(z)\,dz$
Scaling	$x(at),\ a \geq 0$	$\frac{1}{a}X\left(\frac{s}{a}\right)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi j}X_1(s) * X_2(s)$
Initial value	$x(0^+)$	$\lim_{s\to\infty} sX(s) \quad (n > m)$
Final value	$x(\infty)$	$\lim_{s\to 0} sX(s) \quad$ [poles of $sX(s)$ in LHP]

Using Laplace to Solve Differential Equations and System Analysis

- Solving differential equation of third order and higher in the time domain to find the output of the system $y(t)$ is challenging.
- Solving differentials of any order in the Laplace domain is very easy since Laplace transform transfers differential equation into algebraic equation that can be easily solved to find $Y(s)$.
- If you set all initial conditions to zero then you will obtain only the zero-state response of the output $y(t)$.

$$\frac{d^n y}{dt^n} = \frac{d^m x}{dt^m} \rightleftarrows s^n Y(s) - \sum_{k=1}^{n} s^{n-k} \underbrace{\dot{y}^{k-1}(0^-)} = s^m X(s) - \sum_{k=1}^{m} s^{m-k} \underbrace{\dot{x}^{k-1}(0^-)}$$

$$s^n Y(s) = s^m X(s) \longrightarrow Y(s) = \frac{s^m}{s^n} X(s)$$

Initial Conditions

Example

Find the transfer function $H(s)$ of an LTIC system described by the equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

and the system response $y(t)$ if the input $x(t) = 3e^{-5t}u(t)$ and all the initial conditions are zero; that is the system is in the zero state (relaxed).

Answer :

$$y(t) = (-2e^{-5t} - e^{-2t} + 3e^{-3t})u(t)$$

Example: System analysis in Laplace domain

Solve the differential equation to find $y(t)$

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = \frac{dx}{dt} + x(t)$$

Given that $y(0^-) = 2$ and $\dot{y}(0^-) = 1$ and the input $x(t) = e^{-4t}u(t)$

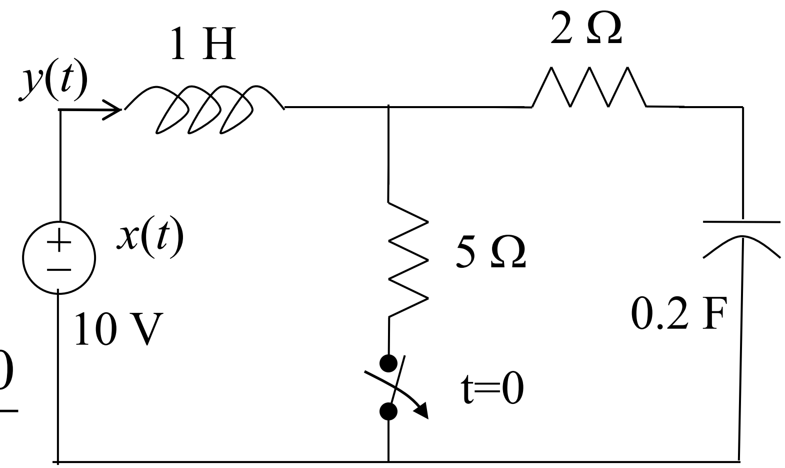
Make sure to show the zero-input response, the zero-state response, and the transfer function of the system $H(s)$.

Answer:

$$y(t) = \underbrace{\left(7e^{-2t} - 5e^{-3t}\right)u(t)}_{\text{zero-input response}} + \underbrace{\left(-0.5e^{-2t} + 2e^{-3t} - 1.5e^{-4t}\right)u(t)}_{\text{zero-state response}}$$

Example

In the circuit, the switch is in the closed position for a long time before $t = 0$, when it is opened instantaneously. Find the inductor current $y(t)$ for $t \geq 0$.



$$L\frac{dy}{dt} + Ry(t) + \frac{1}{C}\int_{-\infty}^{t} y(\tau)d\tau = x(t)$$

$$sY(s) - y(0^-) + 2Y(s) + \frac{5Y(s)}{s} + \frac{5\int_{-\infty}^{0^-} y(\tau)d\tau}{s} = \frac{10}{s}$$

$$y(0^-) = \frac{10}{5} = 2A \qquad \int_{-\infty}^{0^-} y(\tau)d\tau = q_c(0^-) = CV = 2$$

$$sY(s) - y(0^-) + 2Y(s) + \frac{5Y(s)}{s} + \frac{10}{s} = \frac{10}{s}$$

$$y(t) = \sqrt{5}e^{-t}\cos(2t + 26.6^o)u(t)$$

Note:

$i(t)$ = dq/dt

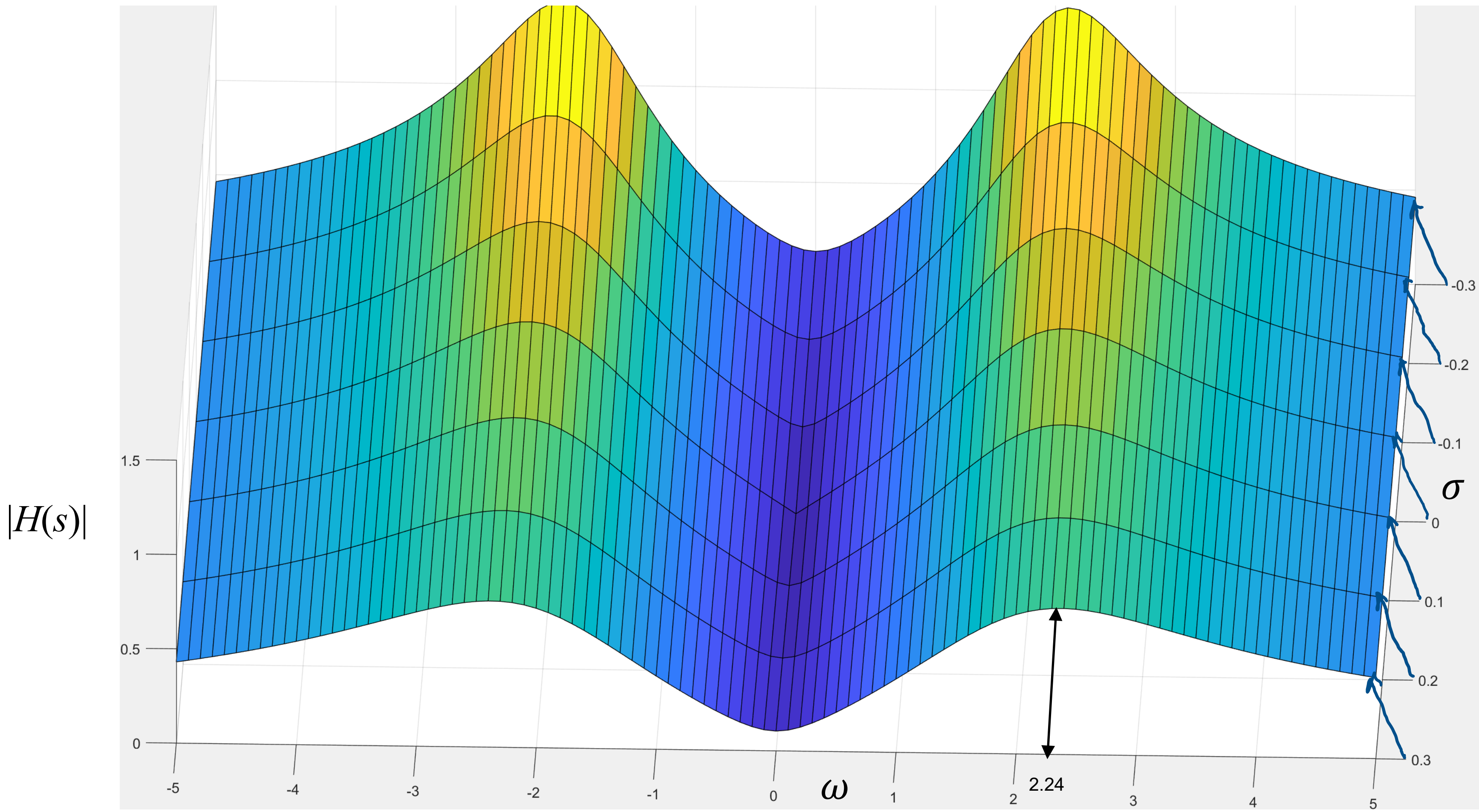
Q = CV

Frequency Response of a LTI System

Frequency Response of a LTI System

Transfer Function $\qquad s = \sigma + j\omega \qquad$ Frequency Response

$$H(s) = \frac{2s}{s^2 + 2s + 5} \xrightarrow{\text{If } \sigma = 0} H(j\omega) = \frac{2j\omega}{-\omega^2 + 2j\omega + 5}$$



Laplace and Frequency Response of a LTI System

$$x(t) = \sum_i X(s_i)e^{s_i t} \qquad\qquad y(t) = \sum_i H(s_i)X(s_i)e^{s_i t}$$

$$X(s) \longrightarrow \boxed{H(s)} \longrightarrow Y(s)$$

$$s = \sigma + j\omega$$

If $\sigma = 0$ in the complex frequency s, so $s = j\omega$ then system analysis is simplified further and it becomes frequency response.

$$x(t) = \sum_i X(j\omega_i)e^{j\omega_i t} \qquad\qquad y(t) = \sum_i H(j\omega_i)X(j\omega_i)e^{j\omega_i t}$$

$$X(j\omega) \longrightarrow \boxed{H(j\omega)} \longrightarrow Y(j\omega)$$

$$x(t) = \mathrm{A}\cos(\omega_o t + 45^o)$$

$$y(t) = \underbrace{|H(j\omega_o)|}_{\text{Amplitude Response}} \mathrm{A}\cos[\omega_o t + 45^o + \underbrace{\angle H(j\omega_o)}_{\text{Phase Response}}]$$

Example: Frequency Response

For the system described by the transfer function

$$H(s) = \frac{s + 0.1}{s + 5}$$

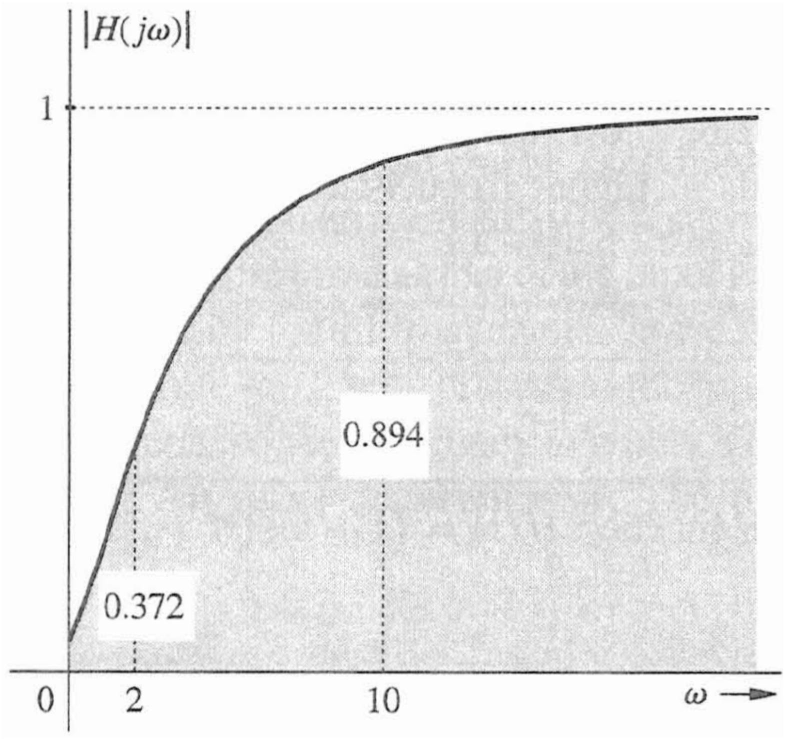
Find the frequency response $H(j\omega)$, and the system response $y(t)$ for input $x(t) = cos2t$ and $x(t) = cos(10t\text{-}50^\circ)$

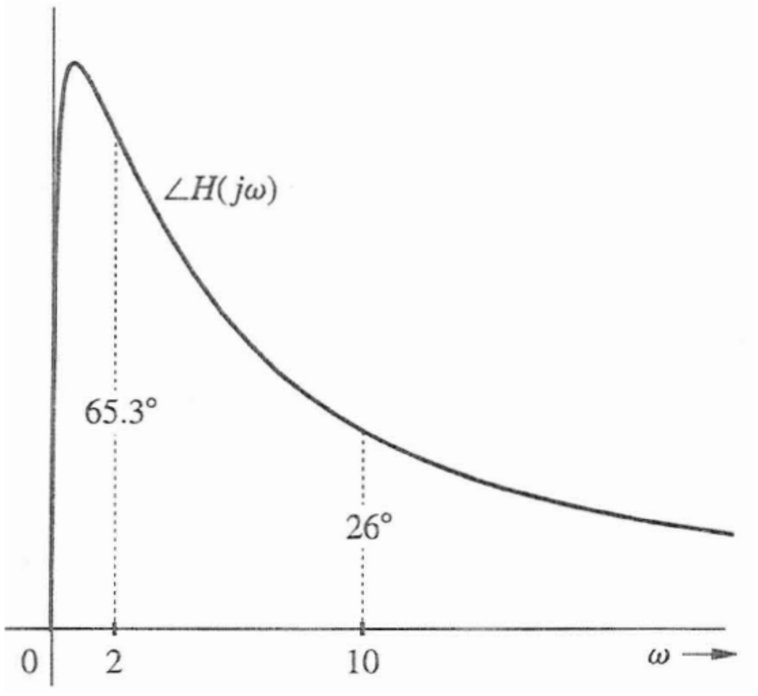
Replace s by $j\omega$

$$H(j\omega) = \frac{j\omega + 0.1}{j\omega + 5}$$

Magnitude and Phase of the Frequency Response

$$\left|H(j\omega)\right| = \frac{\sqrt{\omega^2 + 0.01}}{\sqrt{\omega^2 + 25}} \qquad \Phi(j\omega) = \angle H(j\omega) = \tan^{-1}\left(\frac{\omega}{0.1}\right) - \tan^{-1}\left(\frac{\omega}{5}\right)$$



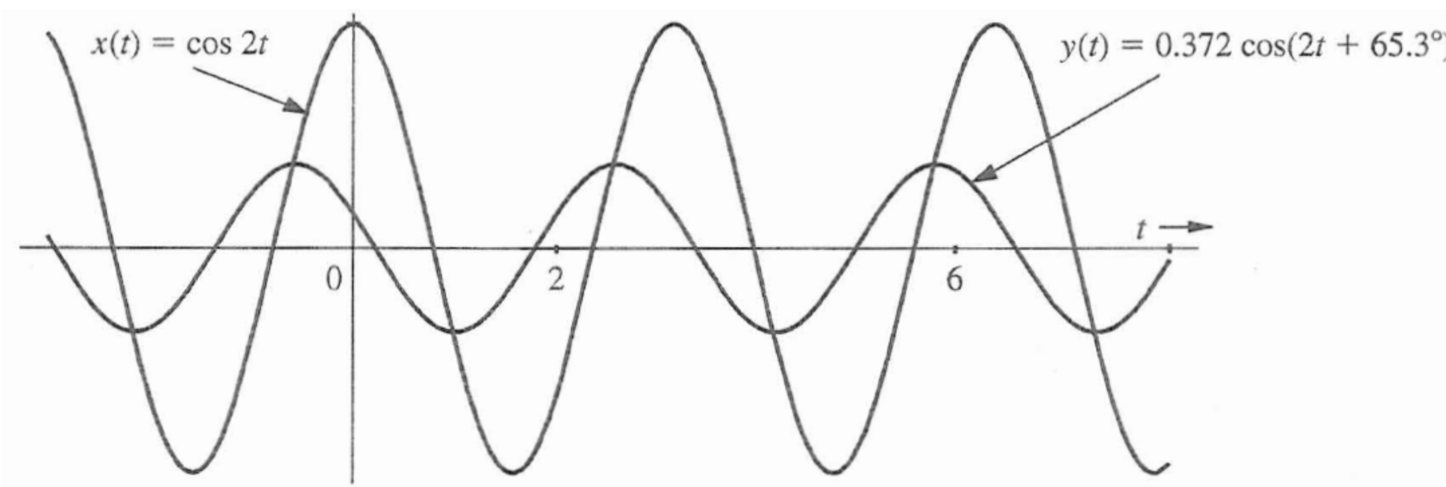


Example: Frequency Response

For input $x(t)=cos2t$, we have:

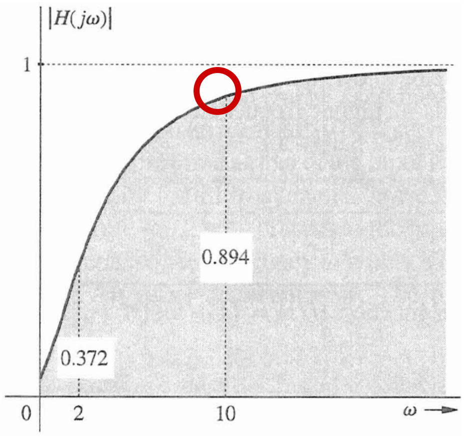
$$|H(j2)| = \frac{\sqrt{2^2 + 0.01}}{\sqrt{2^2 + 25}} = 0.372 \qquad \Phi(j2) = \tan^{-1}\left(\frac{2}{0.1}\right) - \tan^{-1}\left(\frac{2}{5}\right) = 65.3^\circ$$

Therefore $$y(t) = 0.372\cos(2t + 65.3^\circ)$$



Example: Frequency Response

For input $x(t) = \cos(10t - 50°)$, we will use the amplitude and phase response curves directly:

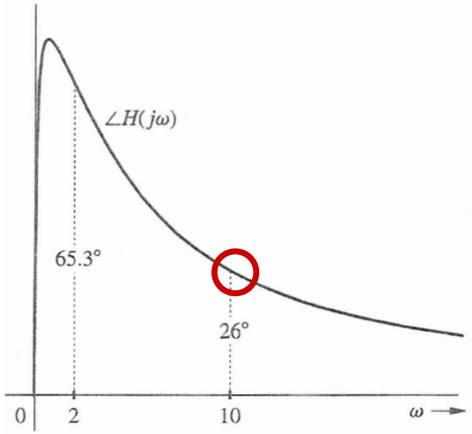


$$|H(j10)| = 0.894$$

$$\Phi(j10) = \angle H(j10) = 26^\circ$$

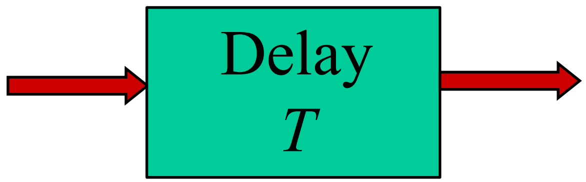
Therefore

$$y(t) = 0.894\cos(10t - 50° + 26°) = 0.894\cos(10t - 24°)$$

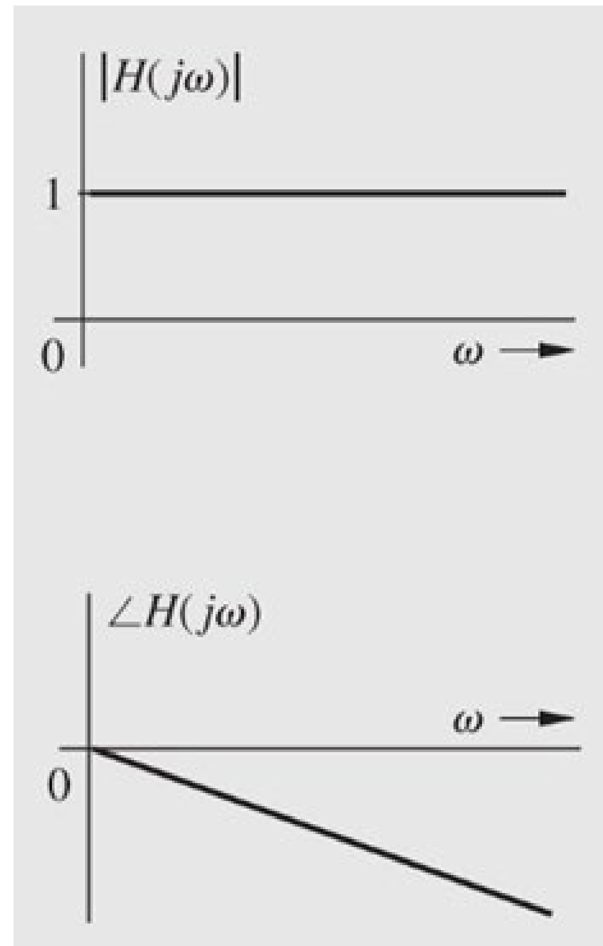


Frequency Response of Ideal Delay

$x(t)$
$X(s)$



$y(t)=x(t-T)$
$Y(s)= X(s)\, e^{-sT}$



$H(s)$ of an ideal T sec delay is:

$$H(s) = e^{-sT} \qquad H(j\omega) = e^{-j\omega T}$$

The magnitude $|H(j\omega)| = 1$

The phase $\theta(j\omega) = -\omega T$

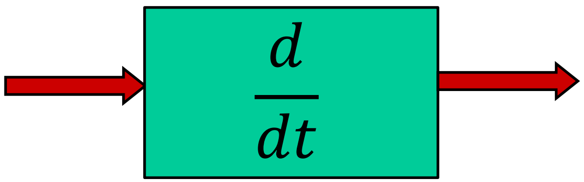
An ideal delay system does not effect the amplitude of the input but phase shift the input with $-T$ gradient.

The group delay is the time delay of all frequencies $\frac{d\theta}{d\omega} = -T$

Frequency Response of an Ideal Differentiator

$x(t)$

$X(s)$



$y(t) = \frac{dx}{dt}$ $\qquad$ $Y(s) = sX(s)$

H(s) of an ideal differentiator is:

$$H(s) = s \qquad\qquad H(j\omega) = j\omega = \omega e^{j\pi/2}$$

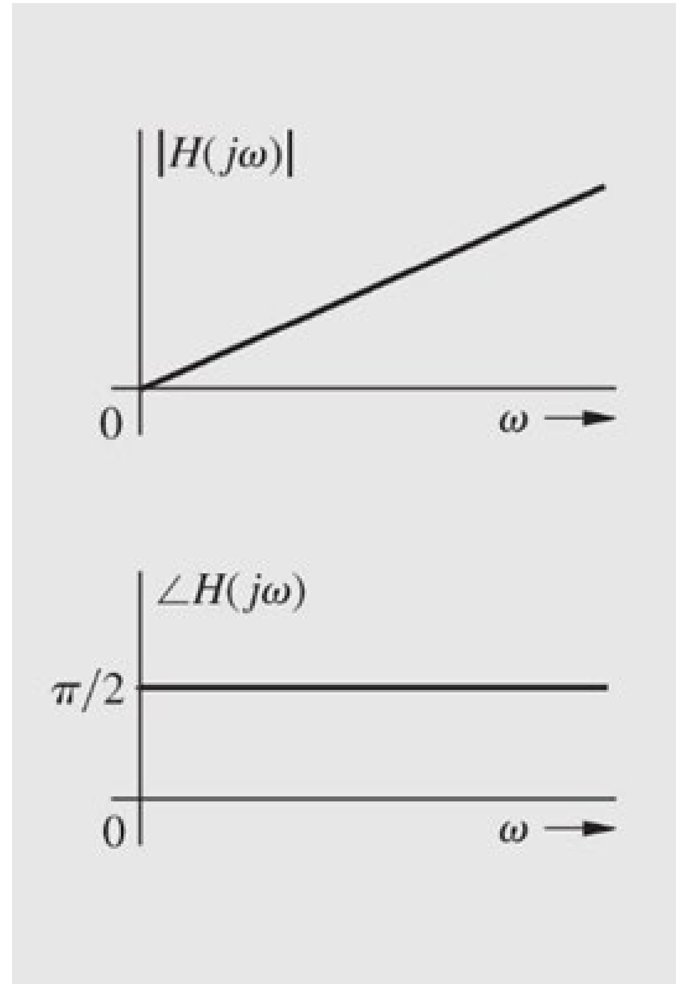
The magnitude $|H(j\omega)| = \omega$

The phase is $\pi/2$

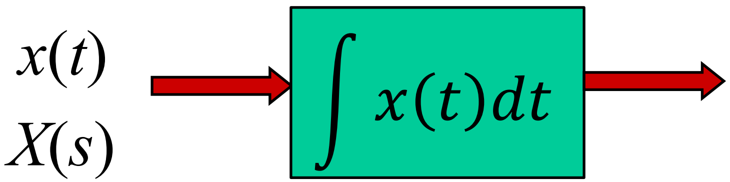
If $x(t) = \cos\omega t$ what is the output

$$\frac{d}{dt}(\cos\omega t) = -\omega\sin\omega t = \omega\cos(\omega t + \pi/2)$$

Differentiators amplify inputs with high frequencies such as noise and for that it is avoided in system design.



Frequency Response of an ideal Integrator



$$y(t) = \int x(t)dt \qquad Y(s) = \frac{1}{s}X(s)$$

H(s) of an ideal integrator is:

$$H(s) = \frac{1}{s} \qquad H(j\omega) = \frac{1}{j\omega} = \frac{1}{\omega}e^{-j\pi/2}$$

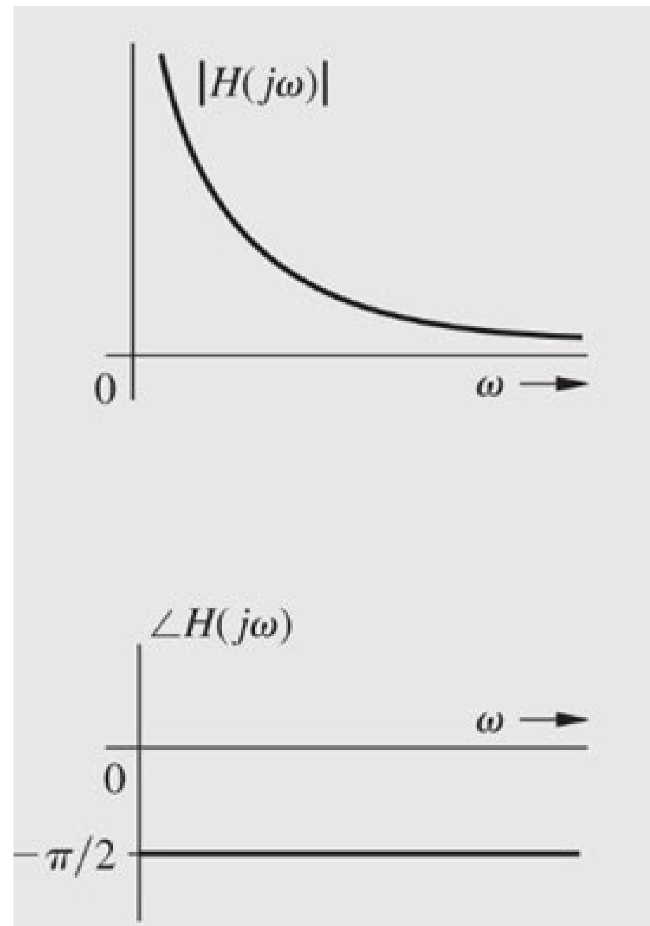
The magnitude $|H(j\omega)| = 1/\omega$

The phase is $-\pi/2$

If $x(t) = \cos\omega t$ what is the output

$$\int \cos\omega t \, dt = \frac{1}{\omega}\sin\omega t = \frac{1}{\omega}\cos(\omega t - \pi/2)$$

Integrator supress inputs with high frequencies such as noise and for that it is used in system design.



Internal Stability

- Internal Stability (Asymptotic)
  - If and only if all the poles are in the LHP
  - Unstable if, and only if, one or both of the following conditions exist:
    - At least one pole is in the RHP
    - There are repeated poles on the imaginary axis
  - Marginally stable if, and only if, there are no poles in the RHP, and there are some unrepeated poles on the imaginary axis.

External Stability BIBO

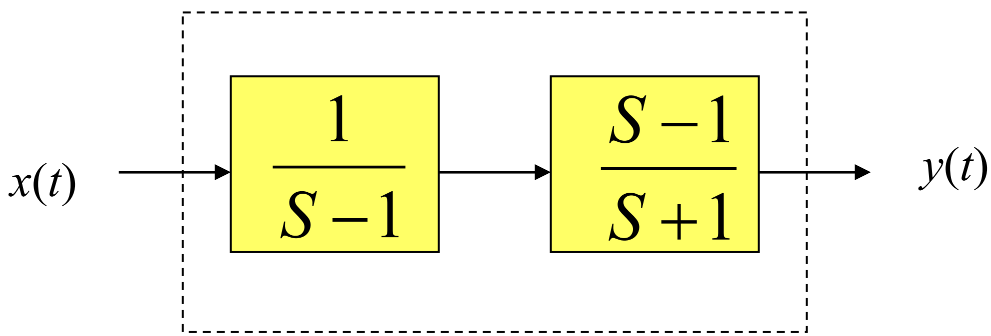
The transfer function $H(s)$ can only indicate the external stability of the system BIBO.

$$H(s) = \frac{b_0 s^M + b_1 s^{M-1} + \ldots + b_M}{s^N + a_1 s^{N-1} + \ldots + a_N}$$

BIBO stable if $M \leq N$ and all poles are in the LHP

Example

Is the system below BIBO and asymptotically (internally) stable?



Topics for Test 2

1. Find the Laplace transform $X(s)$ for signal $x(t)$ using the integral
2. Find the Laplace transform $X(s)$ for signal $x(t)$ using the Laplace properties and table
3. Find the signal $x(t)$ by the inverse Laplace transform of $X(s)$ using the partial fraction expansion, Laplace properties, and the table.
4. Given the differential equation of a system, the input $x(t)$, and the initial conditions of the system, you should be able to find the transfer function $H(s)$, the zero-input response, the zero-state response, the forced response, the natural response, or the total response for the given input $x(t)$.
5. Given the system differential equation or the transfer function $H(s)$, you should be able to find the frequency response $H(j\omega)$, its magnitude and phase. If a dc or sinusoidal input is given then you should be able to find the output $y(t)$ and the steady state response.
6. Given the system differential equation or the transfer function $H(s)$, you should be able to find the roots and zeros of the system and determine its external and internal stability.